

Problem

Using MATLAB, synthesize the periodic waveform for which the Fourier trigonometric Fourier series is

$$f(t) = \frac{1}{2} - \frac{4}{\pi^2} \left(\cos t + \frac{1}{9} \cos 3t + \frac{1}{25} \cos 5t + \dots \right)$$

Step-by-step solution

Step 1 of 1

The function $f(t)$ has a DC offset and is even. We use the following MATLAB code to plot $f(t)$. The plot is shown below. If more terms are taken, the curve is clearly indicating a triangular wave shape which is easily represented with just the DC component and three, cosinusoidal terms of the expansion.

```
for n=1:100
    tn(n)=n/10;
    t=n/10;
    y1=cos(t);
    y2=(1/9)*cos(3*t);
    y3=(1/25)*cos(5*t);
    factor=4/(pi*pi);
    y(n)=0.5- factor*(y1+y2+y3);
end
plot(tn,y)
```

